AJAY SHARMA

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Profile

Software Engineer with 3 years of experience in Java, Spring Boot, Hibernate, SQL, and microservices. Skilled in building real-time APIs and backend services using Kafka for messaging. Experienced in working closely with cross-functional teams including clients, business analysts, and testing teams to deliver reliable and optimized solutions. Focused on performance optimization and maintaining high code quality through comprehensive testing.

Professional Experience

Fiserv **Aug 2023 – Present**

Software Engineer

- Developed backend services to fetch real-time currency rates from third-party systems using Java, Spring Boot, and Kafka, reducing average API response time by 45% and supporting millions of transactions daily.
- Built real-time APIs to fetch loyalty points and personalized offers, increasing offer delivery accuracy by 35% and improving customer engagement metrics.
- Created high-performance queries and analytics APIs to track redirect attempts, reducing report generation time by 40% and enabling actionable insights for business teams.
- Integrated legacy COBOL systems with cloud-based microservices using Java and Spring Boot, reducing downtime incidents by 60%.
- Led migration of COBOL modules to Java-based microservices, enhancing maintainability and deployment agility, resulting in a 50% reduction in deployment time and 35% fewer post-deployment bugs.
- Integrated external systems using custom solutions in Java, Spring Boot, Microservices, COBOL, and SQL, supporting critical cross-platform financial operations and improving data synchronization speed by 40%.
- Wrote unit and integration tests using JUnit and Mockito, achieving 90% coverage and resolving 70+ transaction-related bugs.
- Supported release readiness for EMEA and Middle-East deployments, contributing to successful performance tuning, configuration validation, and regression testing that decreased deployment issues by 25%.

Fiserv **Mar 2023 – Jul 2023**

Intern

- **Rupay Preprocessing Optimization:** Re-engineered a critical payment processing module to resolve recursive loop errors, reducing failure incidents by 25% and enhancing pipeline reliability. Strengthened COBOL settlement validation protocols to ensure 100% data integrity.
- **Release Engineering:** Managed multi-client release cycles through rigorous integration testing and streamlined bug triage. Resolved over 10 critical defects while consistently maintaining 99.9% SLA compliance.
- Designed and coded the frontend website of LiteUI, which is used by banks to check the transaction status and other parameters.

Technical Skills

Languages: Java, COBOL, SQL, Python

Frameworks: Spring Boot, Hibernate-JPA

Messaging: Apache Kafka

Database: MySQL, PostgreSQL

Tools: Git, GitHub, Docker, Maven, Postman, IntelliJ IDEA, Jira, VS Code

Projects

Payment Transaction Processing System | Java, Spring Boot, Kafka, MySQL, Docker **Mar 2026**

- Developed a **microservices-based payment processing system** using Java and Spring Boot to accept, persist, and asynchronously process payment transactions.
- Implemented **Kafka-based event-driven communication** between transaction, payment-processing, and notification services.
- Designed **idempotent payment APIs** to prevent duplicate transaction processing and implemented retry/DLQ mechanisms for failed Kafka messages.
- Used **MySQL with Spring Data JPA** for transaction persistence and designed transaction status workflows for reliable processing.
- Containerized microservices using **Docker Compose** and implemented REST APIs with validation, exception handling, Swagger/OpenAPI documentation, and Actuator monitoring.
- Added **unit and integration tests** using JUnit and Mockito.

Link

Order & Inventory Management System | Java, Spring Boot, Kafka, MySQL, Docker **May 2025**

- Developed a **microservices-based order and inventory management system** using Java and Spring Boot, separating order processing, inventory management, and notification responsibilities.
- Implemented **Kafka-based asynchronous communication** for order creation, inventory reservation, and order-status events.
- Designed an **event-driven inventory reservation workflow** to maintain consistency between orders and available stock.
- Implemented **optimistic locking and idempotent event processing** to handle concurrent inventory updates and duplicate Kafka messages.
- Implemented failure handling using **Kafka retries and Dead Letter Topics (DLT)** for unsuccessful event processing.
- Containerized the application using **Docker Compose** and exposed REST APIs with validation, centralized exception handling, Swagger/OpenAPI documentation, and Actuator monitoring.
- Added **unit and integration tests** for core order, inventory, and event-processing workflows.

Link

Education

Lovely Professional University **Apr 2019 – Jul 2023**

B.Tech Computer Science (CGPA: 8.4) *Punjab, India*

Achievements

- **STAR OF THE MONTH (2 times):** Recognised twice for engineering initiative — for building an idempotency-enforcement layer and an automated duplicate-detection service.